

Percentage, Ratio & Profit/Loss — Level 2 (Multi-Step)

A dedicated harder follow-on to Week 5, for students who've qualified Level 1. This whole topic is new versus Class 4, and Level 2 leans hard on it — chained percentage changes, combined ratios, and profit/loss problems with an extra step. 15 questions, 4-option MCQ, untimed.

Q1. A number is first increased by 25%, then the result is decreased by 20%. If the final value is 600, find the original number. A) 550 B) 575 C) 600 D) 625

Q2. Three numbers are in the ratio 3:4:7. The difference between the largest and smallest numbers is 48. Find the sum of the three numbers. A) 154 B) 161 C) 168 D) 175

Q3. If $A:B = 2:3$ and $B:C = 4:5$, find $A:C$. A) 2:5 B) 8:15 C) 6:10 D) 4:5

Q4. A shopkeeper marks an item 40% above its cost price of ₹500, then sells it after a discount of 15% on the marked price. Find his profit percentage. A) 15% B) 17% C) 19% D) 21%

Q5. A sum amounts to ₹5,200 in 3 years and to ₹6,000 in 5 years at simple interest. Find the rate of interest per annum. A) 8% B) 9% C) 10% D) 12%

Q6. Find 20% of 30% of 5,000. A) 200 B) 250 C) 300 D) 350

Q7. Two numbers are in the ratio 3:5. If 10 is added to each number, the new ratio becomes 2:3. Find the two original numbers. A) 24, 40 B) 27, 45 C) 30, 50 D) 33, 55

Q8. A man sells an article for ₹800, gaining an amount that is 15% of the selling price. Find the cost price of the article. A) ₹660 B) ₹670 C) ₹680 D) ₹690

Q9. In how many years will a sum of money become 4 times itself at 15% per annum simple interest? A) 15 years B) 18 years C) 20 years D) 24 years

Q10. 40% of a number is 25% more than 96. Find the number. A) 280 B) 290 C) 300 D) 310

Q11. An article is sold for ₹456 at a loss of 5%. Find its cost price. A) ₹470 B) ₹475 C) ₹480 D) ₹485

Q12. The ages of two brothers are in the ratio 4:5. After 6 years, the ratio of their ages will be 5:6. Find their present ages. A) 20, 25 B) 22, 27 C) 24, 30 D) 28, 35

Q13. Find the simple interest on ₹8,000 at 9% per annum for 2 years 6 months. A) ₹1,700 B) ₹1,750 C) ₹1,800 D) ₹1,850

Q14. A number is increased by 30%, and then the result is increased by a further 20%. Find the overall percentage increase. A) 50% B) 52% C) 54% D) 56%

Q15. A man buys a bicycle for ₹2,400 and spends ₹300 on repairs. He then sells it at a profit of 20% on his total cost. Find the selling price. A) ₹3,120 B) ₹3,180 C) ₹3,240 D) ₹3,300

Answer key: `answer-key.md`
